

HAO CUI

+1 236-412-5586 | decubingexpert@gmail.com | haocui.dev | linkedin.com/in/hao-cui | github.com/HaoCui

TECHNICAL SKILLS

Languages: Python, C++, Java, Kotlin, TypeScript, JavaScript, HTML, CSS, Bash

Frameworks: React, Next.js, Node.js, Flask, Expo, React Native, Pandas, Tailwind

Technologies: Git, Docker, Google Cloud, Firebase, Nginx, OpenCV, TensorFlow, Pytorch, ONNX, CUDA, MongoDB

EXPERIENCE

Magic Hour (YC W24)

Software Engineer Intern

Nov 2025 - Present

- Built a production-ready z-image turbo generation workflow **used by millions of users** and ran **500+ controlled test renders** to identify the best **sampler, scheduler, and step size**; published results in a clear internal website for reuse.
- Reduced end-to-end generation latency on character replace videos by **49%** by integrating **torch.compile** and **FlashAttention**, plus attention/precision optimizations (**SageAttn2**, **FP16Fast**, **TeaCache**).
- Implemented real-time video processing using **NVIDIA CUDA** and **multithreading** to adapt an open-source multi-GPU GitHub project to run on a **single GPU**; added optimizations including **regex**-based sentence-by-sentence generation splitting and buffering until all frames were generated before display, with large-scale workloads run on **H200** GPUs.

Smashspeed

Chief Technology Officer

May 2025 - Present

- **40k+ downloads across 100+ countries**, amplified by a viral social campaign with **millions** reached.
- Executed an end-to-end port of the machine-learning shuttle tracker to **React Native**.
- Implemented a scalable serverless **Firebase** architecture (Auth, Firestore, Cloud Functions, Cloud Storage).
- Developed the marketing and product website with **React + Tailwind**, sustaining **10k+** monthly active users.
- Helped build the data pipeline using **YOLOv5**, curated and annotated **13k+** images to reach **93% model accuracy**.

FTC Robotics - Furious Frogs

Developer

Sept 2024 - Present

- Built relocalization with an **adaptive Kalman filter**, fusing odometry and camera-based measurements for robust field positioning under match conditions.
- Developed **ball tracking** and autonomous systems that helped rank the team **#1 in Canada** in the 2025-2026 season, peaking at **#2 in the world**.
- Implemented an auto-aim pipeline using **AprilTag detection** with **OpenCV** and tuning PID controllers for fast, stable targeting and shot alignment.

PROJECTS

Steve the Freakysaur (160,000+ impressions) |

- Won **1st place out of 124 participants and 50+ projects** in BC's largest high school hackathon.
- A Chrome Dino-style game controlled entirely using tongue detection through real-time facial segmentation.
- Prototyped in **Pygame** before being ported to **Next.js** with TypeScript for web play.

pip-doom |

- Built a PyPI package that launches a playable terminal version of **DOOM during pip install**, by hooking into the Python package build process so the game runs on the user's machine before installation finishes.
- Forced **source builds only** by shipping sdist without prebuilt wheels, ensuring the install-time execution path always triggers.
- Implemented cross-platform terminal control: on **Windows**, injected real-time key events directly into the console input buffer; on **Linux/macOS**, opened `/dev/tty` directly to bypass pip's redirected I/O and capture live keypresses.

Image 2 Desmos |

- Created a tool to convert raster images into parametric equations suitable for Desmos graphing, enabling visual art to be rendered purely via algebraic curves.
- Implemented image processing pipeline: edge detection → vectorization → equation generation, optimised for limited equation count and interactive use within Desmos.

EDUCATION

Port Moody Secondary School

International Baccalaureate (IB) Diploma Programme

Port Moody, BC

Expected Graduation: 2027